SE Lab 7

Fuzzing for Input Validation Bugs

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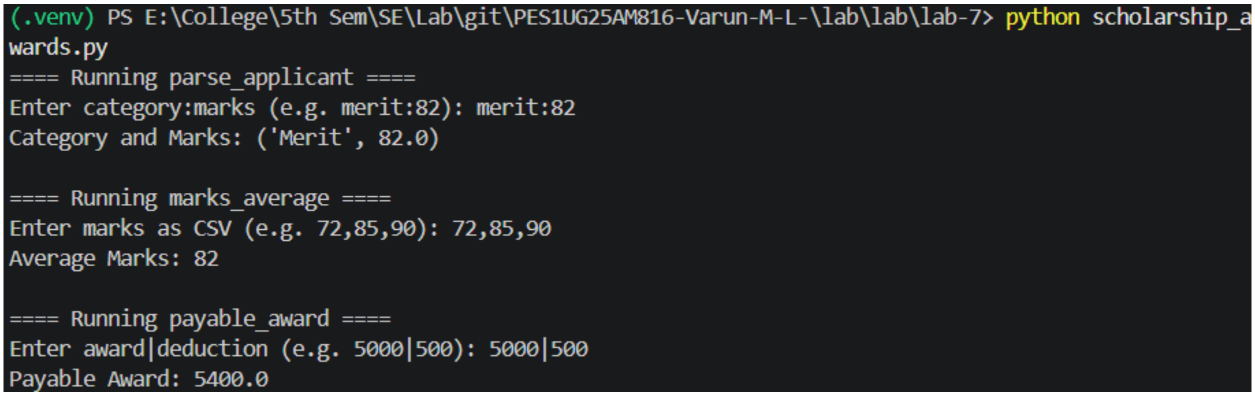
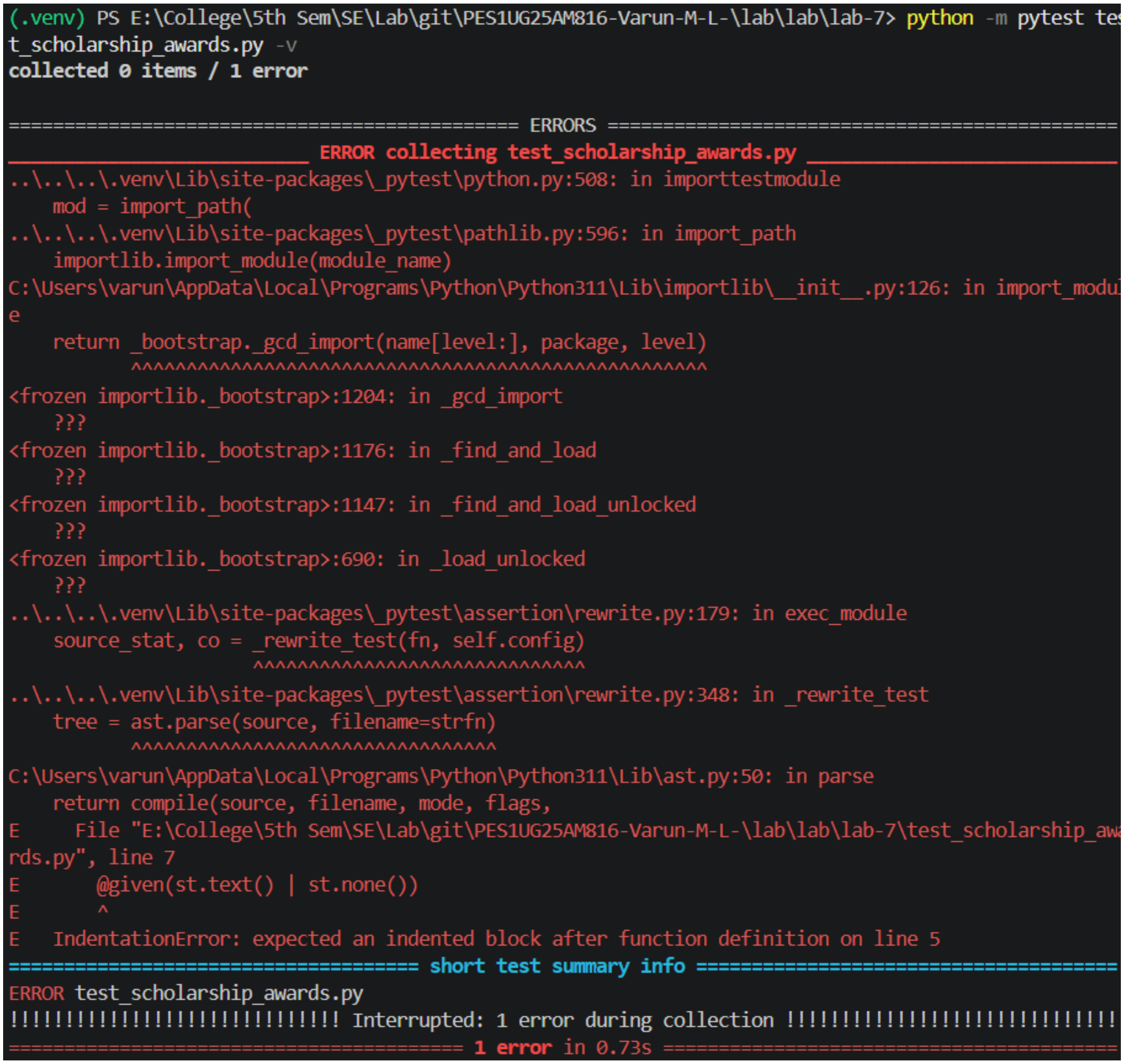
Section: G

Before fixing bugs

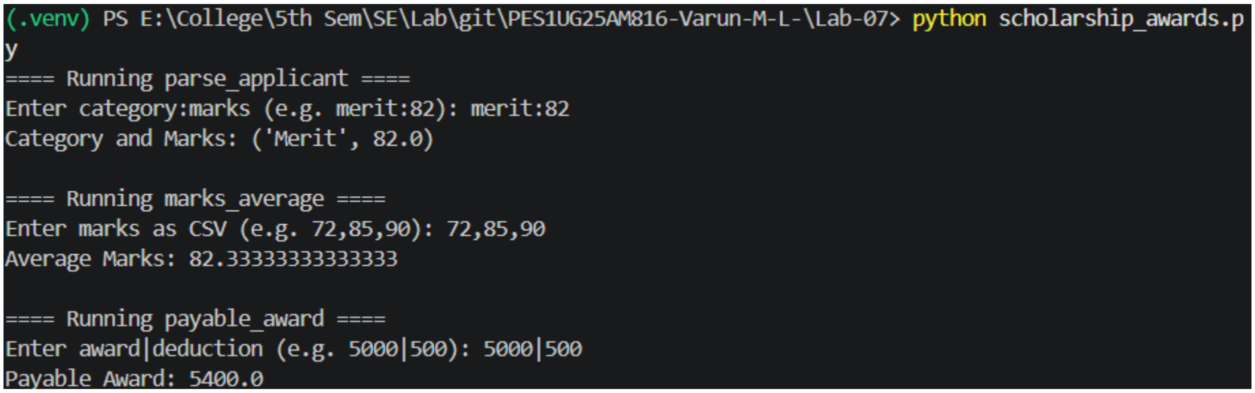


```
(.venv) PS E:\College\5th Sem\SE\Lab\git\PES1UG25AM816-Varun-M-L-\lab\lab\lab-7> python scholarship_a
wards.py
==== Running parse_applicant ====
Enter category:marks (e.g. merit:82): merit:82
Category and Marks: ('Merit', 82.0)

==== Running marks_average ====
Enter marks as CSV (e.g. 72,85,90): 72,85,90
Average Marks: 82

==== Running payable_award ====
Enter award|deduction (e.g. 5000|500): 5000|500
Payable Award: 5400.0
```

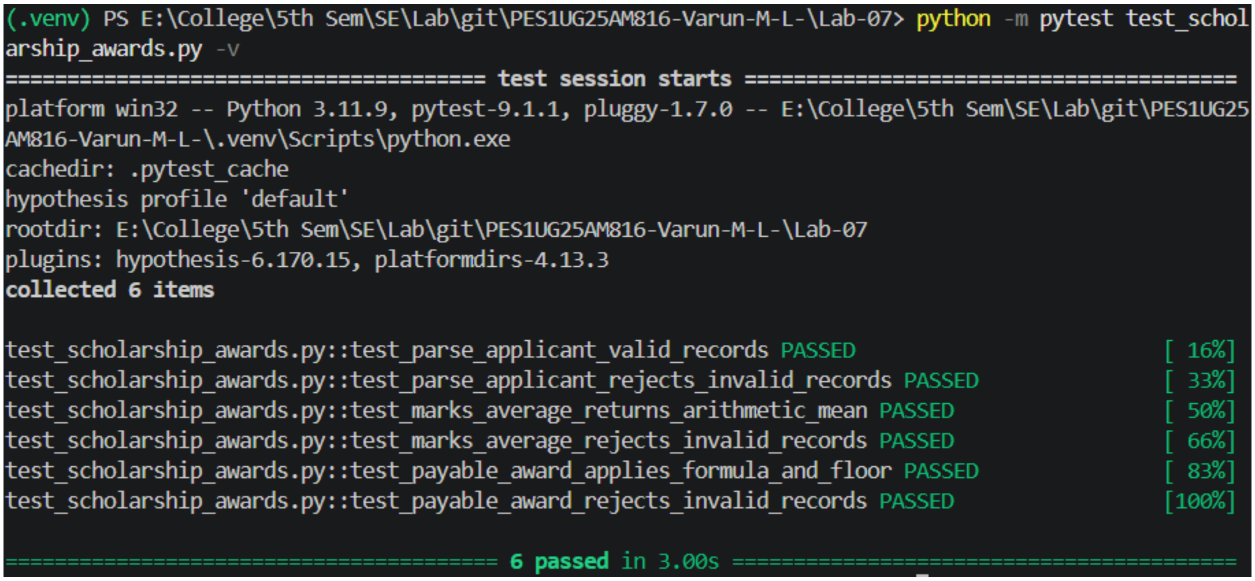


```
(.venv) PS E:\College\5th Sem\SE\Lab\git\PES1UG25AM816-Varun-M-L-\lab\lab\lab-7> python -m pytest tes
t_scholarship_awards.py -v
collected 0 items / 1 error

============================================== ERRORS ==============================================
___________________________ ERROR collecting test_scholarship_awards.py ____________________________
..\..\..\.venv\Lib\site-packages\_pytest\python.py:508: in importtestmodule
    mod = import_path(
..\..\..\.venv\Lib\site-packages\_pytest\pathlib.py:596: in import_path
    importlib.import_module(module_name)
C:\Users\varun\AppData\Local\Programs\Python\Python311\Lib\importlib\__init__.py:126: in import_modu
e
    return _bootstrap._gcd_import(name[level:], package, level)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
<frozen importlib._bootstrap>:1204: in _gcd_import
    ???
<frozen importlib._bootstrap>:1176: in _find_and_load
    ???
<frozen importlib._bootstrap>:1147: in _find_and_load_unlocked
    ???
<frozen importlib._bootstrap>:690: in _load_unlocked
    ???
..\..\..\.venv\Lib\site-packages\_pytest\assertion\rewrite.py:179: in exec_module
    source_stat, co = _rewrite_test(fn, self.config)
                      ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
..\..\..\.venv\Lib\site-packages\_pytest\assertion\rewrite.py:348: in _rewrite_test
    tree = ast.parse(source, filename=strfn)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
C:\Users\varun\AppData\Local\Programs\Python\Python311\Lib\ast.py:50: in parse
    return compile(source, filename, mode, flags,
E     File "E:\College\5th Sem\SE\Lab\git\PES1UG25AM816-Varun-M-L-\lab\lab\lab-7\test_scholarship_aw
rds.py", line 7
E       @given(st.text() | st.none())
E       ^
E   IndentationError: expected an indented block after function definition on line 5
===================================== short test summary info ======================================
ERROR test_scholarship_awards.py
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!! Interrupted: 1 error during collection !!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
========================================= 1 error in 0.73s =========================================
```

After fixing bugs



```
(.venv) PS E:\College\5th Sem\SE\Lab\git\PES1UG25AM816-Varun-M-L-\Lab-07> python scholarship_awards.p
y
==== Running parse_applicant ====
Enter category:marks (e.g. merit:82): merit:82
Category and Marks: ('Merit', 82.0)

==== Running marks_average ====
Enter marks as CSV (e.g. 72,85,90): 72,85,90
Average Marks: 82.33333333333333

==== Running payable_award ====
Enter award|deduction (e.g. 5000|500): 5000|500
Payable Award: 5400.0
```



```
(.venv) PS E:\College\5th Sem\SE\Lab\git\PES1UG25AM816-Varun-M-L-\Lab-07> python -m pytest test_schol
arship_awards.py -v
======================================= test session starts ========================================
platform win32 -- Python 3.11.9, pytest-9.1.1, pluggy-1.7.0 -- E:\College\5th Sem\SE\Lab\git\PES1UG25
AM816-Varun-M-L-\.venv\Scripts\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: E:\College\5th Sem\SE\Lab\git\PES1UG25AM816-Varun-M-L-\Lab-07
plugins: hypothesis-6.170.15, platformdirs-4.13.3
collected 6 items

test_scholarship_awards.py::test_parse_applicant_valid_records PASSED                         [ 16%]
test_scholarship_awards.py::test_parse_applicant_rejects_invalid_records PASSED               [ 33%]
test_scholarship_awards.py::test_marks_average_returns_arithmetic_mean PASSED                 [ 50%]
test_scholarship_awards.py::test_marks_average_rejects_invalid_records PASSED                 [ 66%]
test_scholarship_awards.py::test_payable_award_applies_formula_and_floor PASSED               [ 83%]
test_scholarship_awards.py::test_payable_award_rejects_invalid_records PASSED                 [100%]

======================================== 6 passed in 3.00s =========================================
```